PGCP-AC August 26
Database Assignment No-3

1. Write a query to calculate the total salary of all employees.
   SELECT SUM(Salary) AS TotalSalary
   FROM Employee;

```
mysql> SELECT SUM(Salary) AS TotalSalary
    -> FROM Employee;
+-------------+
| TotalSalary |
+-------------+
|        NULL |
+-------------+
1 row in set (0.01 sec)
```

2. Write a query to find the average salary of employees in each department using GROUP BY.
   SELECT * FROM Employee;
   INSERT INTO Employee
   (EmpID, EmpName, Salary, JoiningDate, DeptID)
   VALUES
   (101, 'Mansi', 49500, '2024-01-10', 10),
   (102, 'Snehal', 55000, '2023-05-15', 20),
   (103, 'Jayesh', 66000, '2022-08-20', 10),
   (104, 'Khushi', 45000, '2024-03-12', 30),
   (105, 'Raj', 70000, '2021-11-05', 20),
   (106, 'Snehal', 52800, '2025-01-15', 10);
   SELECT DeptID, AVG(Salary) AS AverageSalary
   FROM Employee
   GROUP BY DeptID;

```
mysql> SELECT * FROM Employee;
Empty set (0.01 sec)

mysql> INSERT INTO Employee
    -> (EmpID, EmpName, Salary, JoiningDate, DeptID)
    -> VALUES
    -> (101, 'Mansi', 49500, '2024-01-10', 10),
    -> (102, 'Snehal', 55000, '2023-05-15', 20),
    -> (103, 'Jayesh', 66000, '2022-08-20', 10),
    -> (104, 'Khushi', 45000, '2024-03-12', 30),
    -> (105, 'Raj', 70000, '2021-11-05', 20),
    -> (106, 'Snehal', 52800, '2025-01-15', 10);
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT DeptID, AVG(Salary) AS AverageSalary
    -> FROM Employee
    -> GROUP BY DeptID;
+--------+---------------+
| DeptID | AverageSalary |
+--------+---------------+
|     10 |  56100.000000 |
|     20 |  62500.000000 |
|     30 |  45000.000000 |
+--------+---------------+
3 rows in set (0.00 sec)
```

3. Write a query to count the total number of employees in each department.
   SELECT DeptID, COUNT(*) AS TotalEmployees
   FROM Employee
   GROUP BY DeptID;

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Employee
    -> GROUP BY DeptID;
+--------+----------------+
| DeptID | TotalEmployees |
+--------+----------------+
|     10 |              3 |
|     20 |              2 |
|     30 |              1 |
+--------+----------------+
3 rows in set (0.00 sec)
```

4. Write a query to display departments having more than 5 employees using HAVING clause.
   SELECT DeptID, COUNT(*) AS TotalEmployees
   FROM Employee
   GROUP BY DeptID

HAVING COUNT(*) > 5;

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Employee
    -> GROUP BY DeptID
    -> HAVING COUNT(*) > 5;
Empty set (0.01 sec)
```

5. Write a query to list distinct department locations from the Dept table.
   SELECT DISTINCT City
   FROM Department;

```
mysql> SELECT DISTINCT City
    -> FROM Department;
+--------+
| City   |
+--------+
| Pune   |
| Mumbai |
| Nagpur |
+--------+
3 rows in set (0.01 sec)
```

6. Write a query to display the highest salary among all employees. Write a query to display all total number of employee in table.
   SELECT MAX(Salary) AS HighestSalary
   FROM Employee;
   SELECT COUNT(*) AS TotalEmployees
   FROM Employee;

```
mysql> SELECT MAX(Salary) AS HighestSalary
    -> FROM Employee;
+---------------+
| HighestSalary |
+---------------+
|      70000.00 |
+---------------+
1 row in set (0.01 sec)

mysql> SELECT COUNT(*) AS TotalEmployees
    -> FROM Employee;
+----------------+
| TotalEmployees |
+----------------+
|              6 |
+----------------+
1 row in set (0.02 sec)
```

7. Write a query to find employees whose name starts with 'A' using LIKE operator.

SELECT *
FROM Employee
WHERE EmpName LIKE 'A%';

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE EmpName LIKE 'A%';
Empty set (0.00 sec)
```

8. Write a query to find employees whose name ends with 'n' using LIKE operator.
   SELECT *
   FROM Employee
   WHERE EmpName LIKE '%n';

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE EmpName LIKE '%n';
Empty set (0.00 sec)
```

9. Write a query to find employees whose name contains 'ra' using LIKE operator.
   SELECT *
   FROM Employee
   WHERE EmpName LIKE '%ra%';

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE EmpName LIKE '%ra%';
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
+-------+---------+----------+-------------+--------+-------+
1 row in set (0.00 sec)
```

10. Write a query to display all employees sorted by their Salary in descending order.
    SELECT *
    FROM Employee
    ORDER BY Salary DESC;

```
mysql> SELECT *
    -> FROM Employee
    -> ORDER BY Salary DESC;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
|   103 | Jayesh  | 66000.00 | 2022-08-20  |     10 | NULL  |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 | NULL  |
|   106 | Snehal  | 52800.00 | 2025-01-15  |     10 | NULL  |
|   101 | Mansi   | 49500.00 | 2024-01-10  |     10 | NULL  |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 | NULL  |
+-------+---------+----------+-------------+--------+-------+
6 rows in set (0.00 sec)
```

11. Write a query to display all employees sorted by DeptID ascending and then Salary descending.
    SELECT *
    FROM Employee
    ORDER BY DeptID ASC, Salary DESC;

```
mysql> SELECT *
    -> FROM Employee
    -> ORDER BY DeptID ASC, Salary DESC;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   103 | Jayesh  | 66000.00 | 2022-08-20  |     10 | NULL  |
|   106 | Snehal  | 52800.00 | 2025-01-15  |     10 | NULL  |
|   101 | Mansi   | 49500.00 | 2024-01-10  |     10 | NULL  |
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 | NULL  |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 | NULL  |
+-------+---------+----------+-------------+--------+-------+
6 rows in set (0.00 sec)
```

12. Write a query to find employees whose salary is between 30,000 and 60,000.
    SELECT *
    FROM Employee
    WHERE Salary BETWEEN 30000 AND 60000;

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE Salary BETWEEN 30000 AND 60000;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   101 | Mansi   | 49500.00 | 2024-01-10  |     10 | NULL  |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 | NULL  |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 | NULL  |
|   106 | Snehal  | 52800.00 | 2025-01-15  |     10 | NULL  |
+-------+---------+----------+-------------+--------+-------+
4 rows in set (0.01 sec)
```

13. Write a query to display all employees whose DeptID is in (10, 20, 30).
    SELECT *
    FROM Employee
    WHERE DeptID IN (10, 20, 30);

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE DeptID IN (10, 20, 30);
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   101 | Mansi   | 49500.00 | 2024-01-10  |     10 | NULL  |
|   103 | Jayesh  | 66000.00 | 2022-08-20  |     10 | NULL  |
|   106 | Snehal  | 52800.00 | 2025-01-15  |     10 | NULL  |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 | NULL  |
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 | NULL  |
+-------+---------+----------+-------------+--------+-------+
6 rows in set (0.01 sec)
```

14. Write a query to display Min salary of employee .
    SELECT MIN(Salary) AS MinimumSalary
    FROM Employee;

```
mysql> SELECT MIN(Salary) AS MinimumSalary
    -> FROM Employee;
+---------------+
| MinimumSalary |
+---------------+
|      45000.00 |
+---------------+
1 row in set (0.00 sec)
```

15. Write a query to display employees whose JoiningDate is between '2020-01-01' and '2021-12-31'.
    SELECT *
    FROM Employee

WHERE JoiningDate BETWEEN '2020-01-01' AND '2021-12-31';

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE JoiningDate BETWEEN '2020-01-01' AND '2021-12-31';
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
+-------+---------+----------+-------------+--------+-------+
1 row in set (0.00 sec)
```

16. Write a query to display employees whose Salary is NULL.
    SELECT *
    FROM Employee
    WHERE Salary IS NULL;

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE Salary IS NULL;
Empty set (0.00 sec)
```

17. Write a query to display employees whose Salary is NOT NULL.
    SELECT *
    FROM Employee
    WHERE Salary IS NOT NULL;

```
mysql> SELECT *
    -> FROM Employee
    -> WHERE Salary IS NOT NULL;
+-------+---------+----------+-------------+--------+-------+
| EmpID | EmpName | Salary   | JoiningDate | DeptID | Email |
+-------+---------+----------+-------------+--------+-------+
|   101 | Mansi   | 49500.00 | 2024-01-10  |     10 | NULL  |
|   102 | Snehal  | 55000.00 | 2023-05-15  |     20 | NULL  |
|   103 | Jayesh  | 66000.00 | 2022-08-20  |     10 | NULL  |
|   104 | Khushi  | 45000.00 | 2024-03-12  |     30 | NULL  |
|   105 | Raj     | 70000.00 | 2021-11-05  |     20 | NULL  |
|   106 | Snehal  | 52800.00 | 2025-01-15  |     10 | NULL  |
+-------+---------+----------+-------------+--------+-------+
6 rows in set (0.00 sec)
```

18. Write a query to calculate the total salary per department, but only for departments where total salary is greater than 1,00,000 (use HAVING).
    SELECT DeptID, SUM(Salary) AS TotalSalary
    FROM Employee
    GROUP BY DeptID
    HAVING SUM(Salary) > 100000;

```
mysql> SELECT DeptID, SUM(Salary) AS TotalSalary
    -> FROM Employee
    -> GROUP BY DeptID
    -> HAVING SUM(Salary) > 100000;
+--------+-------------+
| DeptID | TotalSalary |
+--------+-------------+
|     10 |   168300.00 |
|     20 |   125000.00 |
+--------+-------------+
2 rows in set (0.00 sec)
```

19. Write a query to display all distinct employee names.
    SELECT DISTINCT EmpName
    FROM Employee;

```
mysql> SELECT DISTINCT EmpName
    -> FROM Employee;
+---------+
| EmpName |
+---------+
| Mansi   |
| Snehal  |
| Jayesh  |
| Khushi  |
| Raj     |
+---------+
5 rows in set (0.00 sec)
```

20. Write a query to count the number of departments having the same location.
    SELECT City, COUNT(*) AS TotalDepartments
    FROM Department
    GROUP BY City
    HAVING COUNT(*) > 1;

```
mysql> SELECT City, COUNT(*) AS TotalDepartments
    -> FROM Department
    -> GROUP BY City
    -> HAVING COUNT(*) > 1;
Empty set (0.00 sec)
```